

Assessing bird response patterns using flight initiation distance: evidence from multiple urban park surveys in Nanjing, China

Zhewen Ye^a, Qian Xiong^a, Yao Xiong^b, Yunfeng Yang^{a,*} 

^a College of Landscape Architecture, Nanjing Forestry University, Nanjing 210037, China

^b College of Art and Design, Nanjing Forestry University, Nanjing 210037, China

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ABSTRACT

Urban parks not only fulfill multiple functions within urban ecosystems but also serve as key sites for frequent interactions between wildlife and humans, thereby creating complex and dynamic symbiotic environments. As one of the most representative animal groups, birds' behavioral patterns reflect both their adaptation to urban environments and the influence of urban habitats on wildlife spatial use. However, empirical studies on how environmental characteristics shape birds' alert behavior remain limited. We used flight initiation distance (FID) as a quantitative behavioral indicator, assessing birds' sensitivity to human approach in eight urban parks in Nanjing, China. The results showed that certain bird species exhibited relatively high tolerance and became dominant by continually adapting to human activities, whereas non-dominant species tended to avoid disturbances. Trees emerged as a critical factor affecting FID across species, particularly in areas with low vegetation cover. Compared with long-term stress, instantaneous disturbance exerted broader and more consistent impacts in parks across different urban development gradients. These findings highlight the notable potential of FID in urban ecological management and planning. Prioritizing the conservation of non-dominant species may help improve the ecological security of urban bird populations. Mitigating adverse impacts through optimized spatial zoning and enhanced vegetation conditions can further support this goal. This study offers valuable guidance for the conservation of bird diversity and the refinement of habitat design in urban parks.

1. Introduction

With the continuous expansion of urban areas, the boundary between nature and human activity has become increasingly blurred. Urban parks, as typical forms of green infrastructure, serve as unique spaces shared by humans and wildlife (Jenkins et al., 2021; Kostzewa and Krauze-Gryz, 2020). These parks provide relatively stable shelter and breeding habitats for wildlife and play an essential role in urban biodiversity conservation. The presence of wild animals also enhances the ecological services offered by parks, enriching residents' experiences with nature and increasing public awareness and concern for environmental issues (Duncan et al., 2019; Mutanga et al., 2017). Therefore, park managers should strike a balance between ecological value and recreational function, ensuring the safety of wildlife while reasonably allocating space for human use (Barcelos et al., 2022).

However, the establishment of such a symbiotic relationship is marked by strong species selectivity and spatial constraints, as environmental changes require birds to continuously adapt to shifting

conditions (Heggie-Gracie et al., 2020; Srinivasan et al., 2019). Some bird species that are more tolerant of human disturbances exhibit spatial use patterns that overlap with human activities. In contrast, disturbance-sensitive species are often disadvantaged in resource and space competition, struggle to maintain stable populations, and may face elevated survival risks or even gradually disappear from urban areas due to chronic disturbances. This disparity in adaptability can lead to biotic homogenization in urban habitats (Joyce et al., 2018; Liang et al., 2019), ultimately undermining species diversity and ecosystem functionality (Kale et al., 2018).

Birds are among the most successful urban adapters, and the status of urban bird populations has become a vital indicator of urban ecological quality (Snep et al., 2016). In recent years, researchers have used life history traits and vigilance behavior to assess the tolerance of birds across varying flock sizes, body sizes, and locations. Birds may adapt to urban environments by altering phenotypes or behaviors, such as shifting activity timing (Spelt et al., 2021), increasing vocalization frequency (Walters et al., 2019), or improving foraging flexibility (Chard

* Corresponding author.

E-mail address: yangyf@njfu.edu.cn (Y. Yang).

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et al., 2018). Their responses to human presence significantly influence habitat selection and adaptation. Therefore, understanding these behavioral shifts is crucial for addressing the ecological challenges posed by human-driven landscape changes.

A critical metric for assessing birds' sensitivity to potential threats is the flight initiation distance (FID), the distance at which a bird takes flight in response to an approaching human under standardized conditions. FID offers a direct and quantifiable measure of birds' risk assessment and escape decision-making (Loeffler-Henry et al., 2021). In urban settings, human activities influence bird behavior through various forms, leading to complex human-bird interactions (Feng and Liang, 2020). For example, artificial disturbances such as drones and vehicles have been shown to trigger flight responses (DeVault et al., 2017; Jarrett et al., 2020; Schlacher et al., 2013), while repeated feeding can alter foraging strategies and significantly reduce birds' wariness toward humans (Derryberry and Coomes, 2020; Plummer et al., 2019). However, most existing studies have focused on the short-term effects of isolated disturbances, often targeting individual species or specific green spaces. This narrow scope limits our understanding of interspecific behavioral variation at the community level and constrains insights into behavioral adaptation under multi-source disturbance scenarios.

Numerous studies have explored the relationship between urbanization levels and avian flight initiation behaviors (delBarco-Trillo, 2018; Lin et al., 2012), generally highlighting birds' behavioral responses to urban-rural gradients (Charutha et al., 2021; Tätté et al., 2018). Nevertheless, such studies often overlook the substantial environmental heterogeneity within cities. The escape strategies of birds are strongly influenced by local habitat features, such as vegetation height and canopy cover, which directly affect their responses to threat (Campbell, 2016). Given the considerable variability among urban parks in vegetation structure, human presence, and management intensity, mechanisms underlying bird behavioral shifts may differ across parks (Morelli et al., 2018). Yet research on bird FID variation among different urban green spaces remains limited.

To address this gap, we conducted field surveys in eight urban parks in Nanjing to investigate birds' flight initiation responses. Our goal was to evaluate how urban environmental characteristics influence FID. We also examined whether birds adjust their escape strategies in response to external conditions, thereby reflecting their behavioral plasticity and adaptability to urban habitats. Based on this framework, we proposed three hypotheses: (1) Bird FID serves as an effective indicator of their behavioral responses to disturbance levels and habitat quality in urban parks; (2) The relationship between habitat variables and bird escape behavior varies across parks with different levels of urban development; (3) Different species adopt distinct escape strategies in response to potential threats. Through this study, we aim to deepen the understanding of the behavioral ecology of urban birds and provide a scientific reference for ecological protection and disturbance management in urban green spaces.

2. Methods and materials

2.1. Study area

Nanjing (31°14' N–32°37' N, 118°22' E–119°14' E), located in eastern China, served as the study area from October 2022 to July 2024. The terrain of Nanjing is characterized by an interlaced distribution of low mountains, hills, and plains. Hills are mainly distributed in the eastern and southern parts of the city, while the plains stretch along both banks of the Yangtze River. The elevation of the region ranges from 1.8 to 448.9 m, with an average altitude of 20 to 30 m. Nanjing experiences a northern subtropical humid monsoon climate, marked by distinct seasons and abundant rainfall. Spring and autumn are relatively short, while summer and winter are comparatively longer, resulting in a pronounced annual temperature range. The region's natural vegetation primarily consists of a mixed forest of deciduous and evergreen broad-

leaved species. Located in the ecological transition zone of the middle and lower reaches of the Yangtze River, Nanjing lies along the East Asia–Australasia migratory bird route, making it an ideal site for bird research.

Eight representative urban parks were selected as survey sites along the radial direction extending outward from the city center of Nanjing, covering the central urban area, suburban zones, and peripheral regions (Fig. 1). These parks have a relatively consistent functional composition, typically including typical urban green space elements such as woodlands, water bodies, and artificial structures. Given differences in park size and shape, one to two transects were established in each park to capture a wide range of habitat types. To ensure spatial independence of the data, the distance between transects was set at a minimum of 200 m. The width of the transect was determined by the range on both sides where individual birds and their behaviors could be clearly observed. To avoid duplicate records of the same bird individual, all surveys proceeded in a single direction along each transect without turning back. Surveys were carried out twice monthly, resulting in at least 20 observations in total for each park.

2.2. Measurement of FID

Field observations were conducted in weather conditions free of precipitation or strong winds, during periods of abundant natural light, specifically from two hours after sunrise to two hours before sunset. Observers approached birds on foot to simulate predator encounters, and the birds' flight initiation responses were then observed and recorded (Blumstein, 2006). Specifically, wearing non-brightly colored clothing to minimize disturbance (Zhou and Liang, 2020), researchers approached target individuals slowly along a straight path at a constant speed of 0.5 m/s. When a bird abruptly altered its previous activity pattern and took off directly, the observer immediately stopped and remained still. A laser rangefinder was then used to measure the distance between the observer and the bird's take-off location, which was recorded as the FID (Fig. 2c). Compared to alert distance (AD), FID has clearer criteria and is less influenced by subjective observer judgment. Therefore, FID was selected as the primary behavioral response variable in this study. In the case of bird flocks, we chose the individual closest to the observer as the focal subject (Zhao et al., 2024a). Only individuals that had not shown obvious responses to human presence and were not located within 100 m of any predator were tested (Van Donselaar et al., 2018). Bird identification primarily followed "A Field Guide to the Birds of China", with individual recognition based on external characteristics such as body size and plumage coloration to prevent repeated recordings of the same individual (Carlen et al., 2021).

2.3. Measurement of environmental variables

During field investigations, each time an individual bird was observed, its location was designated as a sampling point. The distance between any two points was maintained at no less than 50 m (Ekanayake et al., 2022). Habitat assessments primarily consisted of evaluating surrounding environmental conditions and plant community characteristics. Environmental surveys focused on the intensity of human disturbance at each point, while vegetation assessments encompassed multiple plant-related variables (Table 1), including tree-layer species richness (TRS), shrub-layer species richness (SRS), herbaceous-layer species richness (HRS), horizontal species richness of the plant community (HSR), vertical structural complexity of the plant community (VSC), and vegetation coverage (VC). Centered on each sampling point, a quadrat was established to measure the vegetation variables around the observed individual. Due to variations in habitat conditions and topography across sites, the quadrat size ranged from 50 m × 40 m to 50 m × 50 m, ensuring consistent sampling intensity within each unit area of green space (Uchida et al., 2021).

Due to the numerous small-scale quadrats involved in this study,

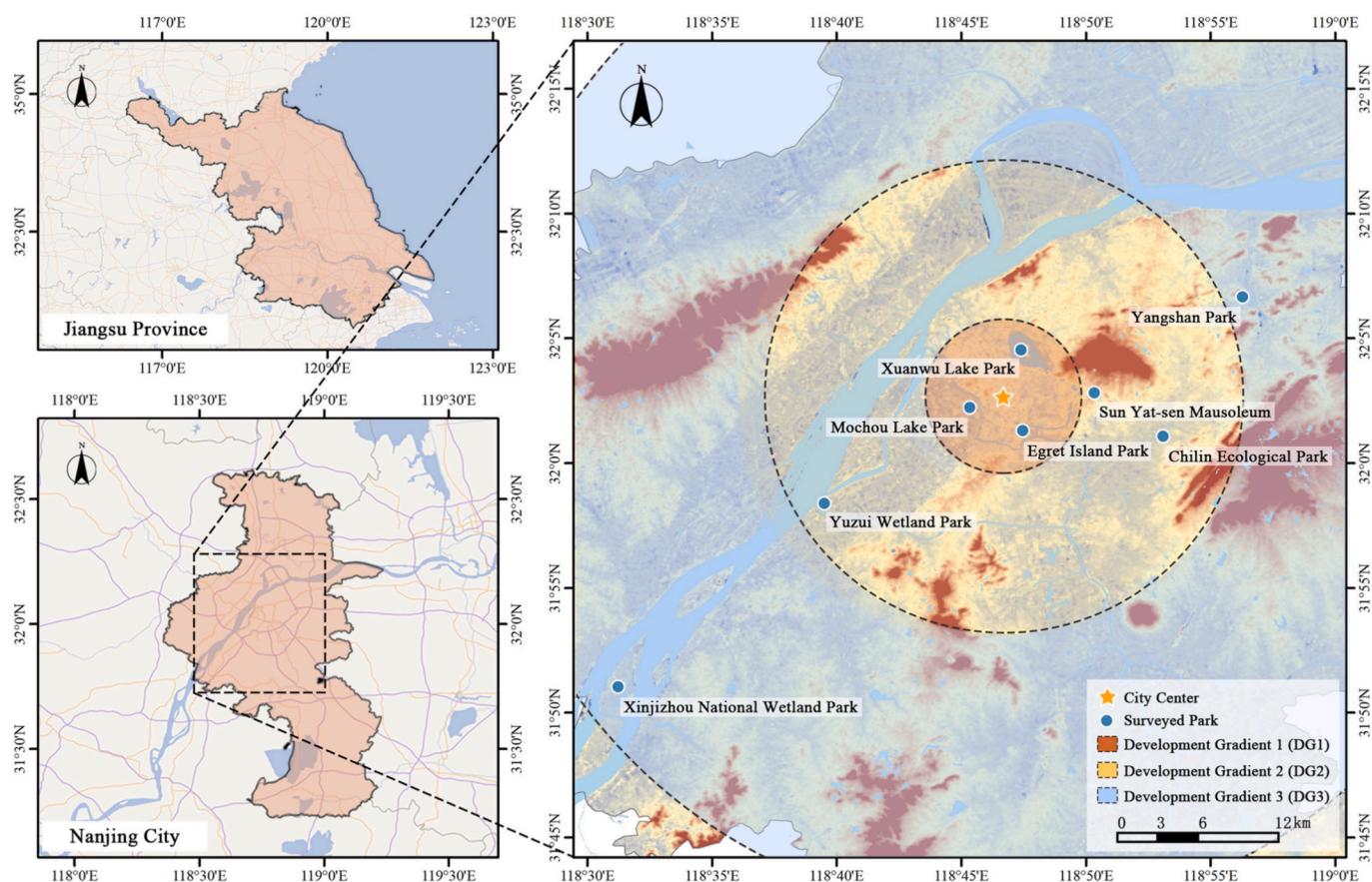


Fig. 1. Location map of the eight urban park sites in Nanjing.

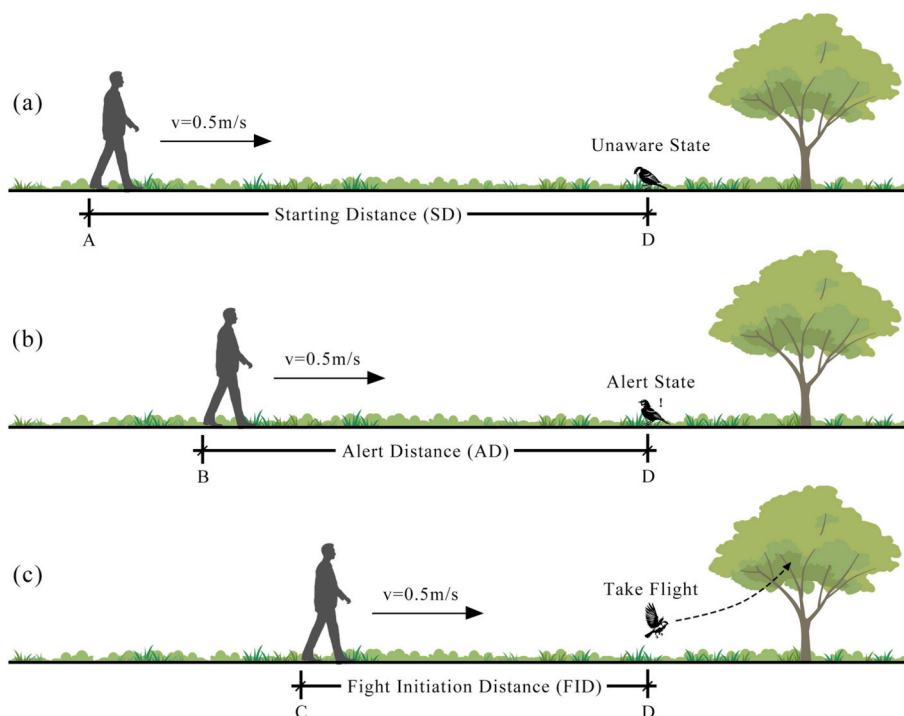


Fig. 2. Diagram of the FID measurement procedure. The observer begins at position A and approaches the target individual located at position D. Upon reaching position B, the target individual exhibits alert behavior, such as turning its head or looking at the observer, signaling the detection of a potential threat. As the observer continues to approach position C, the target individual ultimately decides to flee.

Table 1
Details of each vegetation variable.

Vegetation variables	Description
Tree-layer species richness (TRS)	Number of tree species within the quadrat
Shrub-layer species richness (SRS)	Number of shrub species within the quadrat
Herbaceous-layer species richness (HRS)	Number of herbaceous species within the quadrat
Horizontal species richness of the plant community (HSR)	Number of species present horizontally in the plant community
Vertical structural complexity of the plant community (VSC)	Degree of structural complexity of the plant community in the vertical direction
Vegetation coverage (VC)	Vertical projection area of vegetation (%) within the quadrat

automatic classification methods are prone to errors in such contexts. Therefore, VC was determined using manual interpretation and vectorization. Field survey data were combined with orthophotos obtained from Google Earth to identify vegetation within each quadrat through visual interpretation, followed by manual vectorization in ArcGIS 10.8. To minimize subjective bias during the identification process, all vegetation variables were consistently processed by a single researcher. Furthermore, based on the presence of trees, small trees, shrubs, small shrubs, and herbaceous ground cover, VSC was classified into five levels (Fig. 3). Specifically, Level 1 refers to a single-layer or simplified structure that contains only one type of vegetation, such as trees, shrubs, or herbaceous plants, lacking clear vertical stratification. Level 2 consists of two layers, including trees and herbaceous ground cover. Level 3 includes trees, shrubs, and herbaceous ground cover. Level 4 adds small trees to the previous three layers. Level 5 represents the highest level of complexity and includes all five vegetation components. We assessed human disturbance using two types of indicators. Instantaneous disturbance was captured as the sound level (SL) by measuring the on-site sound pressure level after each bird's startle, using a MicW i436 microphone (MicW Audio, China) and the SPLnFFT Sound Meter v6.2 application. Long-term disturbance intensity was represented by the distance from the sampling point to the nearest park path or road (RdDST). Additionally, the distance from each sampling point to the nearest water body (WaDST) was recorded.

2.4. Statistical analysis

Firstly, we applied the K-means clustering method, using the straight-line distance from each park to the center of Nanjing as an indicator for preliminary classification, and further divided the parks into three categories based on the dendrogram generated by hierarchical clustering. Urbanization indicators within a 500-m circular buffer around each park, including road density, population density, and building density were then calculated (Chen et al., 2024; Shi et al., 2024), with road data obtained from OpenStreetMap. PERMANOVA results revealed significant differences among these categories in the selected indicators, effectively reflecting the actual development status of the parks ($R^2 = 0.88$, $P < 0.05$). Following the “Environmental Quality Standard for Noise”, which defines 55 dB as the acceptable daytime noise threshold for urban park green spaces, we classified bird occurrence points into high and low sound level zones accordingly.

To examine differences in birds' startled flight responses, we applied the Kruskal-Wallis test and the Mann-Whitney *U* test, followed by multiple comparisons with Bonferroni correction. For these non-parametric tests, we calculated the mean and standard error (SE) for each group. In addition, Epsilon-squared was computed to compare multiple groups, while the effect size *r* was calculated for pairwise comparisons. The combined evaluations helped assess the potential impact of sample size differences on the significance of the results. Ultimately, the SEs for each group in the non-parametric tests were relatively small, and the effect sizes for all significant results exceeded the medium effect level, confirming that the results were unlikely to be affected by differences in sample size.

Using the “lme4” package, we constructed generalized linear mixed models (GLMMs) to identify the vegetation variables that explained changes in bird tolerance, modeling each urban development gradient separately. To avoid multicollinearity issues, we performed variance inflation factor (VIF) tests on vegetation variables, and all VIF values were below 10, indicating acceptable collinearity levels. In each model, FID was the response variable, vegetation factors were fixed effects, and bird species were included as random effects to control for interspecific behavioral variability. All continuous explanatory variables were standardized before fitting the models. By incorporating random effects, GLMMs effectively address hierarchical data structures and non-normality.

To prevent the overall average effect from masking differences, we

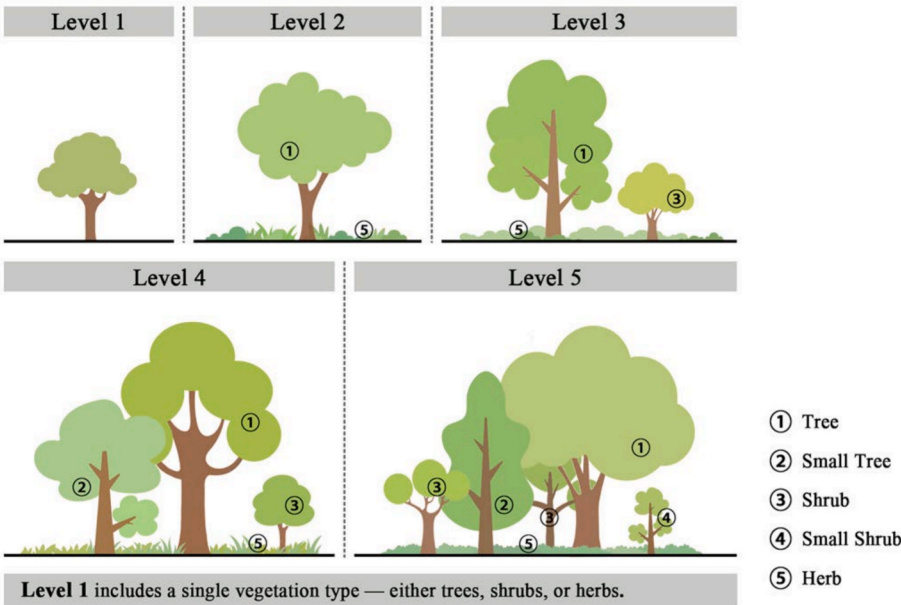


Fig. 3. Classification of vertical structure in plant communities.

compared the performance of dominant and non-dominant bird species. Previous studies have shown that the classification of dominant versus non-dominant species depends on the spatial scale and regional characteristics of the study area (Borges et al., 2019; Mena et al., 2016). Our study found that some species were recorded at least 10 times across different development gradients, with their occurrence frequency reaching 25 % in one or more site types. Species' occurrence frequency not only reflects their spatial distribution and habitat-use stability within the study area, but also indirectly indicates their functional role in the ecological network (Georgiev et al., 2020; Lin et al., 2025). Considering both local high abundance and regional high occurrence frequency, two of the nine recorded species were defined as dominant to reveal species-specific responses to environmental factors. This classification allows us to highlight the role of dominant species within the community while retaining sufficient sample information for non-dominant species, thereby providing a comprehensive view of the adaptive strategies of different species in urban environments.

Multiple linear regression models (MLRs) with interaction terms were employed to analyze behavioral differences of birds under various human disturbances and to examine whether these effects varied across urban development levels. In addition, the "xgboost" package was used to construct regression models for further evaluation of environmental effects on FID and identification of key factors influencing bird FID. Extreme gradient boosting (XGBoost) offers advantages such as automatic modeling of nonlinear relationships and handling of variable interactions. Its built-in regularization mechanism effectively prevents overfitting, while support for parallel computing and efficient feature selection makes it particularly suitable for complex, variable datasets. The dataset was split into training and testing sets at a ratio of 75 % to 25 % to ensure the reliability of model training. During model training, cross-validation was applied, and hyperparameter tuning was conducted using grid search. The optimal model was selected based on minimizing the root mean squared error (RMSE), with lower RMSE values indicating better predictive performance on unseen data. All statistical analyses were performed in R 4.3.1.

3. Results

3.1. Effects of urban development gradient on bird FID

We collected a total of 295 valid bird FID observations. The overall mean FID was 7.39 ± 0.17 m (mean $\pm$ SE), ranging from 2.35 m to 16.66 m, indicating variation in individual responses and their associated environments. DG1, DG2, and DG3 are park sites categorized

through K-means clustering, representing varying levels of urban development, with a distribution from high to low (Fig. 4). Results from the Kruskal-Wallis test and subsequent multiple comparisons revealed significant differences in FID across development gradients ($P < 0.001$). Specifically, the mean FID in DG3 was significantly higher than in DG1 and DG2 ($P < 0.001$), whereas no significant difference was found between DG1 and DG2. The average FID for DG1 was 6.47 ± 0.22 m, for DG2 was 6.77 ± 0.29 m, and for DG3 reached 9.50 ± 0.28 m. FID also exhibited differences between areas with high and low sound levels ($P < 0.001$), particularly in DG1 and DG2 ($P < 0.01$).

3.2. Plasticity and regulation patterns in bird FID

We grouped the sample points based on the vegetation cover within each quadrat to investigate how vegetation conditions influenced bird flight responses. The results showed that TRS had a significant negative effect on birds' FID ($P < 0.001$), suggesting that tree-layer vegetation in urban green spaces may offer better visual cover and escape routes. Further analysis indicated that when VC exceeded 60 %, in addition to TRS, multiple factors including SRS, HRS, and VSC also significantly influenced FID. ($P < 0.05$). In DG3, birds' FID showed relatively low dependence on vegetation. When VC exceeded 30 %, no vegetation variable had a significant effect (Fig. 5). The extent to which vegetation factors affect FID varied across different urban development gradients. Notably, vegetation factors had the strongest explanatory power at DG2, where the GLMM achieved the highest goodness of fit ($R^2 = 0.55$), indicating that birds in this gradient were most sensitive to habitat structure.

To examine differences in birds' sensitivity to human disturbances, we compared FID values of different bird species (Fig. 6). The results revealed that FID was significantly influenced by human disturbance across all sample sites ($P < 0.001$). In DG1, SL was the primary factor affecting the dominant species, significantly reducing their FID ($P < 0.01$). In DG2, SL affected the FID of both dominant and non-dominant species ($P < 0.001$), with a stronger response observed in non-dominant species. By contrast, human disturbances in DG3 had no significant effect on FID, and differences between dominant and non-dominant species were minimal, indicating overall low sensitivity. Furthermore, RdDST had a marginally significant effect on the FID of dominant species in DG1, but reached significance in DG2 ($P < 0.01$), suggesting that dominant species may be more tolerant of road environments than non-dominant species. The regression results indicated that the interaction between environmental variables and site type was significant, suggesting that birds' FID responses to human disturbances varied with the

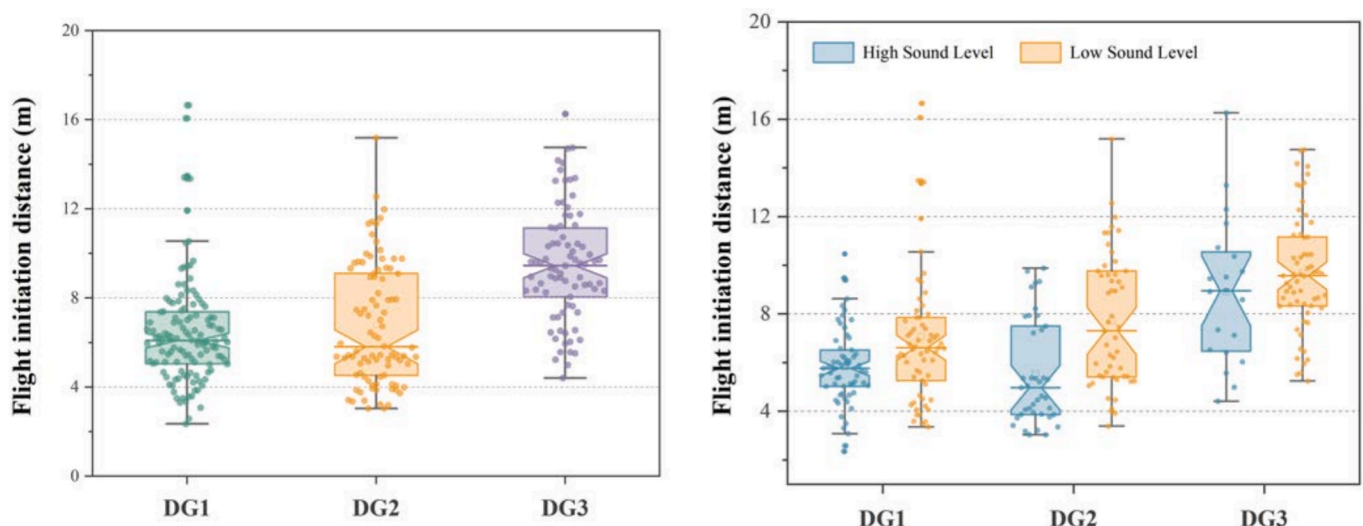


Fig. 4. Comparison of birds' FID under different urban development gradients and sound levels.

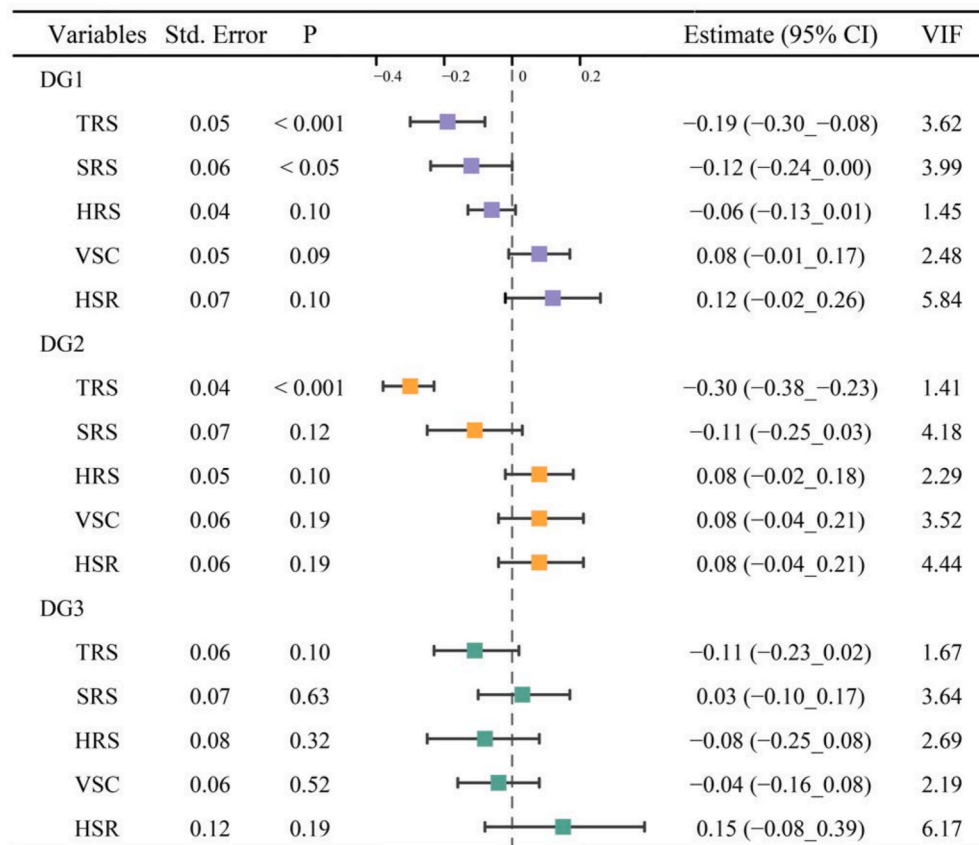


Fig. 5. Effect sizes of vegetation factors on FID in quadrats with vegetation coverage exceeding 30 %. Labels: TRS = tree-layer species richness, SRS = shrub-layer species richness, HRS = herbaceous-layer species richness, VSC = vertical structural complexity of the plant community, HSR = horizontal species richness of the plant community. The squares represent the estimates of each variable, while the line segments indicate their corresponding 95 % confidence intervals (CI). The vertical dotted line at zero (Estimate = 0) serves as a no-effect reference. Squares positioned to the left of the dotted line suggest a potential negative effect, whereas those to the right indicate a positive effect. The standard error (Std. Error) reflects the variability of the estimate, and smaller standard errors correspond to more reliable estimates. Statistical significance is assessed at a p-value (P) threshold of less than 0.05.

degree of urban development.

3.3. Identification of key factors influencing bird FID

The importance of environmental factors for FID was assessed using the XGBoost model (Fig. 7). For dominant species, the DG2 model (RMSE = 1.57, $R^2 = 0.61$, MAE = 1.14) and DG3 model (RMSE = 2.05, $R^2 = 0.60$, MAE = 1.58) both highlighted TRS, SL, and RdDST as the primary drivers of variation. In DG2, VC exerted the strongest influence on the FID of non-dominant species (RMSE = 2.14, $R^2 = 0.80$, MAE = 1.82), whereas in DG3, their FID was shaped mainly by the combined effects of VC and WaDST (RMSE = 2.29, $R^2 = 0.44$, MAE = 1.94). In contrast, in DG1, the FID of dominant (RMSE = 2.46, $R^2 = 0.33$, MAE = 2.15) and non-dominant species (RMSE = 1.72, $R^2 = 0.37$, MAE = 1.15) was highly stochastic and difficult to predict due to environmental disturbances. Overall, the XGBoost results indicated that dominant species maintained relatively stable FID across urban development gradients, whereas non-dominant species were more sensitive to environmental disturbances, showing pronounced shifts in FID and larger fluctuations in model explanatory power.

For the environmental variable model across all samples (RMSE = 2.03, $R^2 = 0.52$, MAE = 1.23), the most important factors were TRS, SL, RdDST, VC, and WaDST. Compared with other variables, SL and TRS had relatively stable effects. Partial dependence curves of FID showed that as SL increased, FID decreased across development gradients for different bird species (Fig. 8). In DG2, the effect of SL on FID was primarily concentrated in low-sound-level areas. For non-dominant species, the

influence of SL was more variable, fluctuating notably between 50 and 55 dB, likely due to differences in the birds' positions within the sites. Even when non-dominant species were sensitive to human disturbances, in areas with transient disturbances, they could mitigate impacts by utilizing surrounding cover. In DG3, FID of dominant species decreased significantly when SL reached 50 and remained stable at higher sound levels. The primary effect of TRS on non-dominant species was observed within the 0–8 range, while dominant species' dependence on tree-layer vegetation was mainly within 0–6.

3.4. Potential effects of functional group differences on bird FID

Beyond comparisons between dominant and non-dominant species, differences among functional groups further reflect behavioral diversity under urban development gradients. Mann-Whitney *U* tests showed that, across all valid samples, migratory birds had significantly higher FID than resident birds ($P < 0.05$). Among recorded birds classified by diet, three species were carnivorous and the remainder omnivorous, with significant FID differences between these groups ($P < 0.05$). Additionally, some species were recorded at least 10 times across DG1–DG3. Further analysis of these species revealed that Rock Pigeon (*Columba livia*) exhibited distinct behavior from others, as its FID did not vary across development gradients ($P > 0.05$).

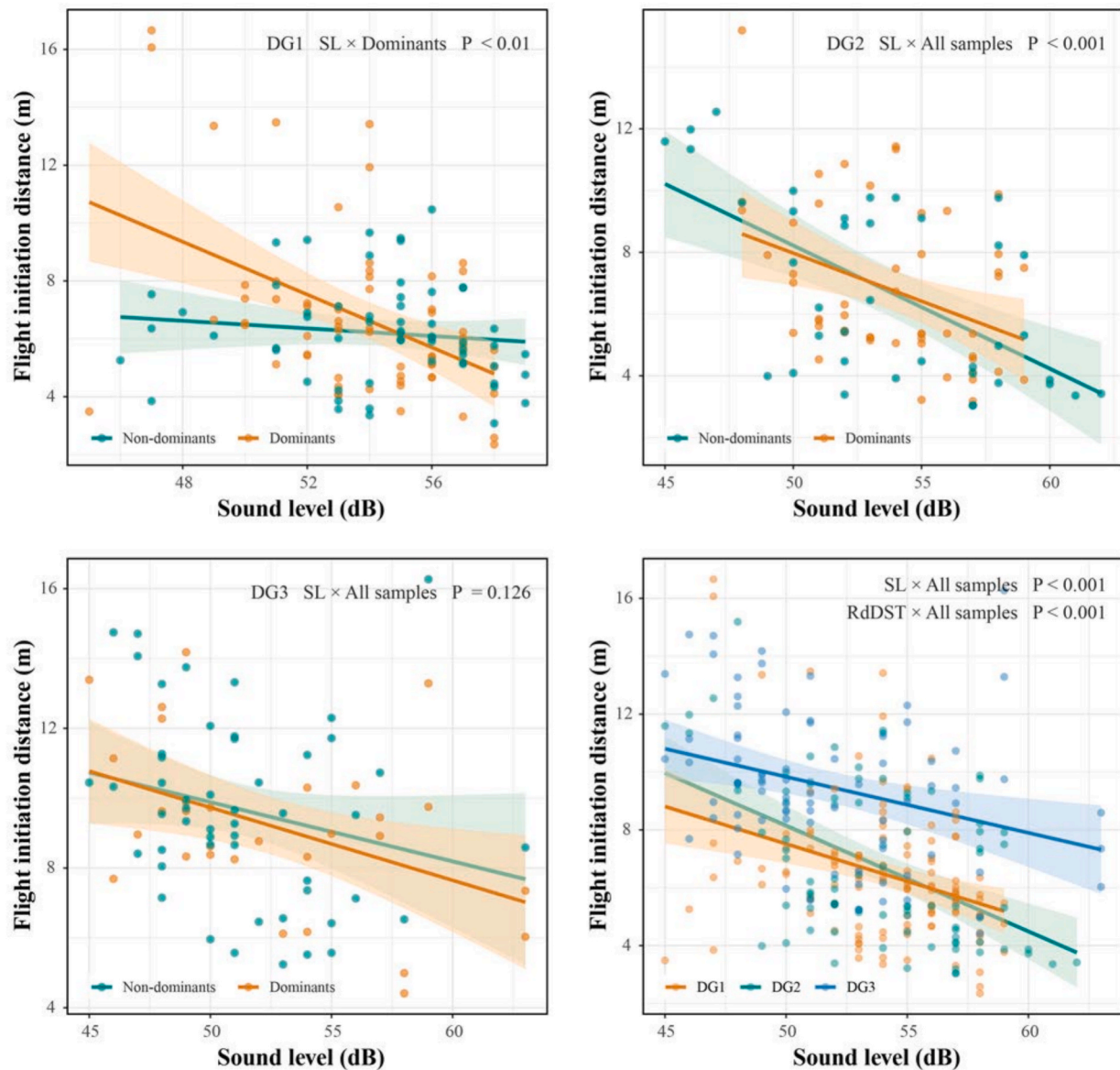


Fig. 6. Relationship between human disturbances and birds' FID. Original data are represented by points. Labels: SL = sound level, RdDST = the distance to the nearest park path or road.

4. Discussion

4.1. Effects of human disturbance on birds' flight initiation behavior

Most studies have shown that urban birds generally exhibit higher tolerance to close human approach compared with rural birds (Symonds et al., 2016; Vincze et al., 2016). Our findings are consistent with previous research, confirming that the urbanization degree has a significant effect on birds' escape behavior (Hall et al., 2020). With increasing levels of urban development, birds became less sensitive to human presence, as reflected in shorter FID. This behavioral shift provides empirical support for the urban gradient response theory. Among the nine variables examined, human disturbances emerged as particularly important factors. Notably, regardless of the degree of urban development, transient human disturbance (sound level) in parks significantly reduced birds' tolerance to human approach, affecting even species typically considered less adaptable (Matyjasiak et al., 2023).

At the same time, within different urban habitats, some individuals remain highly alert while others adopt more exploratory behaviors

(Samia et al., 2017). The prevalence of dominant bird species in cities can be attributed to ecological filtering, whereby obligate and disturbance-sensitive species are gradually excluded, leaving only a few tolerant and behaviorally plastic species to dominate. Except in the least developed sites, the goodness of fit of the XGBoost model was consistently higher for non-dominant than for dominant species. Dominant species typically occupy the core ecological niches of a site and have stricter requirements for habitat stability. Consequently, human interference may at any time compromise the efficiency with which they exploit resources. By contrast, non-dominant species often rely on sub-optimal resources and mitigate disturbances by enhancing the utilization of marginal environments (Sprau and Dingemanse, 2017).

Higher sound levels generally shortened FID, whereas greater distance from paths and roads tended to increase it. In highly disturbed areas, birds may reduce escape distance to minimize energy expenditure and opportunity costs, while in rarely visited areas, they often display heightened vigilance (Thibault et al., 2020). Compared with sound level, the effect of distance from paths and roads on FID proved more variable. To capture the overall bird community within parks, we employed the

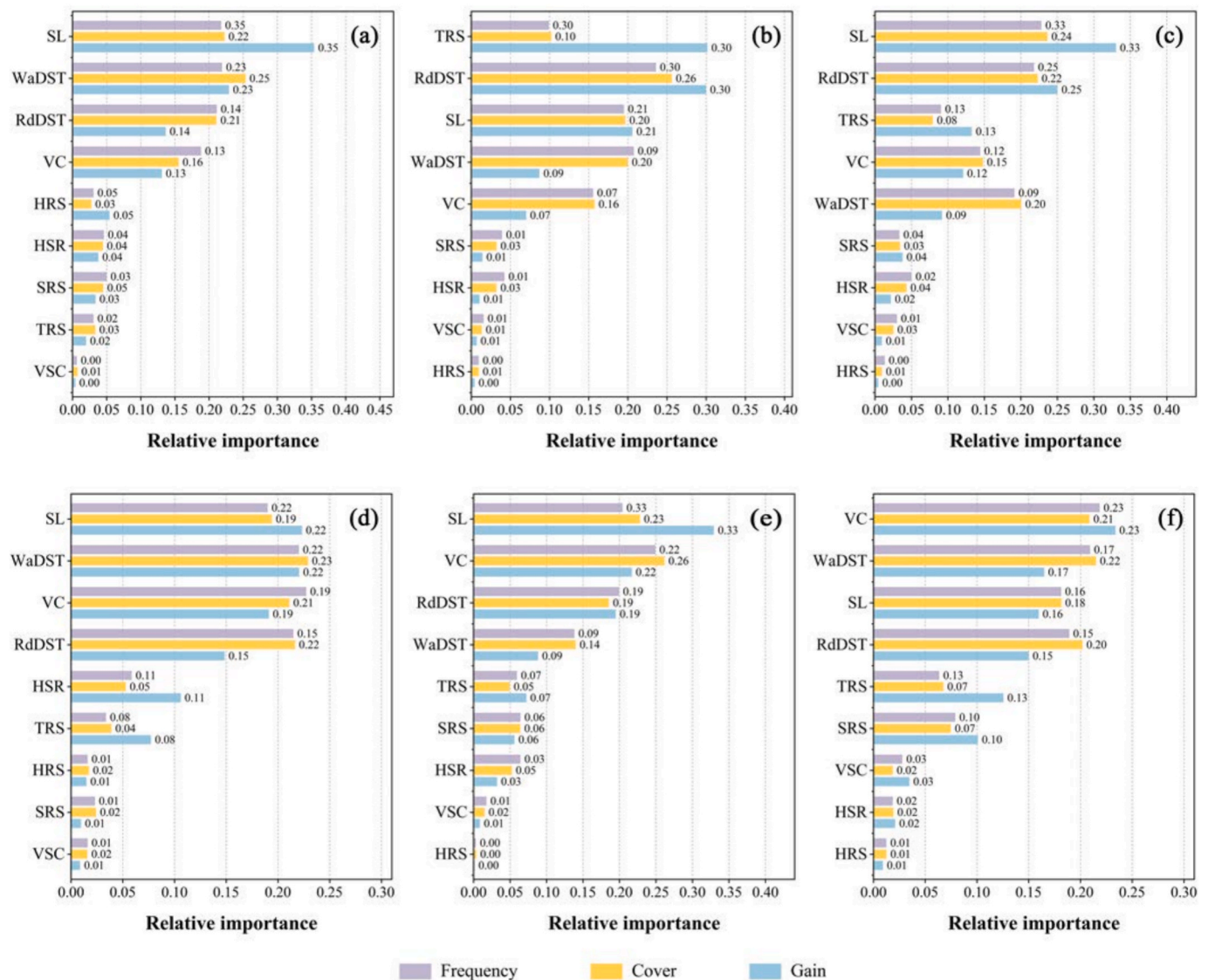


Fig. 7. Relative importance of environmental features for (a) DG1 dominant species, (b) DG2 dominant species, (c) DG3 dominant species, (d) DG1 non-dominant species, (e) DG2 non-dominant species, and (f) DG3 non-dominant species. Labels: TRS = tree-layer species richness, SRS = shrub-layer species richness, HRS = herbaceous-layer species richness, HSR = horizontal species richness of the plant community, VSC = vertical structural complexity of the plant community, VC = vegetation coverage, SL = sound level, WaDST = the distance to the nearest water body, RdDST = the distance to the nearest park path or road. In the XGBoost model, Gain represents the average contribution of a given feature to improving model performance and serves as a key metric for evaluating feature importance. Cover refers to the sum of sample weights associated with a feature when it is used for node splitting, while Frequency indicates how often the feature is used across all tree splits.

transect method, drawing data from areas characterized by diverse landscape features. Such spatial heterogeneity may partly obscure the role of individual environmental variables. Overall, our results suggest that birds' responses to human disturbance are not merely the result of habituation (Gravolin et al., 2014), but more likely reflect their active capacity to cope with external pressures through behavioral choice and experiential learning (Garamszegi and Moller, 2017; Moller and Díaz, 2018; Tryjanowski et al., 2020).

4.2. Regulation of bird escape responses by vegetation structure

We found that within urban parks, trees played a pivotal role in shaping birds' escape behavior. Dense canopies provide vertical escape routes away from human activities, allowing birds to evade threats promptly and thereby potentially shortening their FID (Radvan et al., 2023). In areas with lower vegetation cover, birds may flush earlier due to increased exposure, with more random escape trajectories (Osorio-

Beristain et al., 2018). Compared with deciduous species, evergreen trees retain foliage year-round, offering stable shelter, particularly in winter, and enhancing birds' sense of security (Zhao et al., 2024c). Moreover, tree-rich areas often support bird aggregations, where individuals can use group vigilance to assess potential threats (Morelli et al., 2019). Previous studies have identified trees as the primary refuge for birds when facing disturbance (Morelli et al., 2022b), and our findings further confirm that FID may be linked to the proximity of available tree refuge. Additionally, in densely vegetated areas, the synergistic effects of different plant types significantly influenced FID, likely because birds in resource-rich environments weigh multiple habitat factors when making escape decisions, resulting in significant contributions from several variables.

In parks with the lowest level of development, however, vegetation-driven effects were less evident. Even at sample points with relatively high vegetation cover, multiple vegetation variables had no significant influence on FID. Instead, XGBoost results highlighted distance from

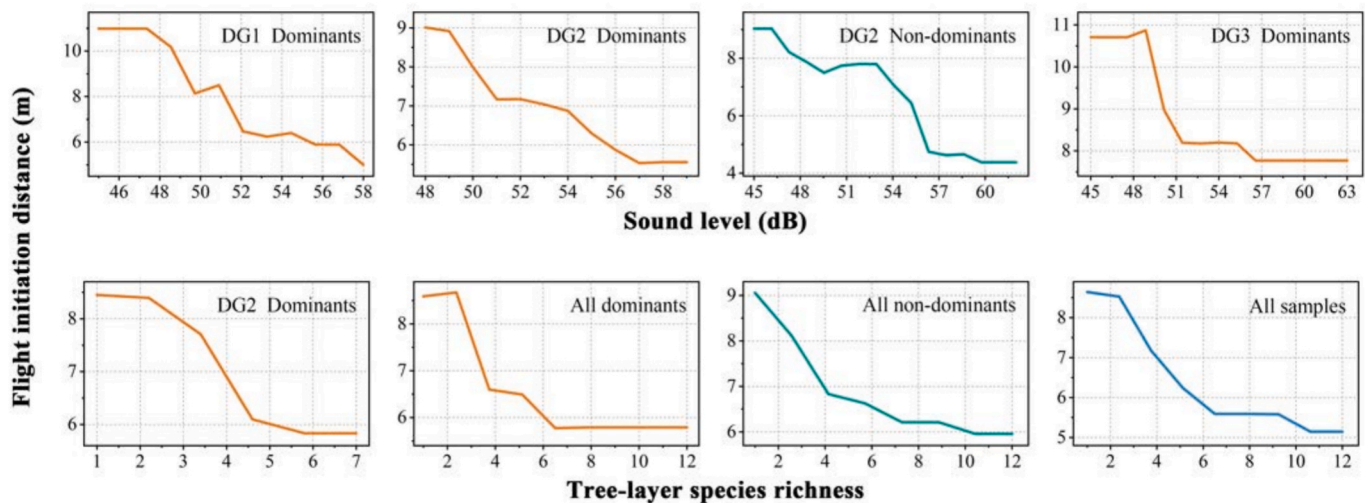


Fig. 8. Partial dependence of key environmental variables on birds' FID.

water bodies as an important predictor of escape behavior in these areas. We suggest that this difference may arise from variations in park greening characteristics and resource distribution. Suburban parks with low development levels are typically large in scale and experience relatively few visitors. Birds in these parks are less restricted, and nearby water bodies, located away from human activity, directly shape their escape behavior (Canedoli et al., 2018). In contrast, parks with higher development gradients are located in the urban core, where visitor density is high. Only more diverse and three-dimensional vegetation structures provide adequate concealment for birds. Our field data further indicated that vegetation conditions in urban parks were positively correlated with the degree of urban development, with lower levels of urbanization linked to poorer vegetation indicators. This pattern likely reflects differences in green space management strategies and landscape aesthetics. Although suburban parks provide larger green areas, the lack of systematic management or ecological design may limit the availability of effective shelter.

4.3. Behavioral ecological implications for urban green space management

In this study, we quantified birds' perception of and behavioral responses to environmental disturbances in parks by their FID, thereby providing insights into their adaptation strategies under different levels of urban development. Consistent with our hypothesis, species differed in their escape strategies, with such variation being particularly pronounced in the parks with the highest level of urban development. Dominant species generally exhibited greater behavioral flexibility, appearing more frequently in areas with high human activity. In contrast, non-dominant species were more vigilant, with more pronounced fluctuations in their FID. Across all urban development gradients, Oriental Magpie (*Pica serica*) was recorded with a frequency exceeding 25 %, and was therefore identified as a dominant species. In the least developed parks, its occurrence frequency reached 37 %. Long-term observations from the China Bird Report Center (<https://www.birdreport.cn/>) also identify Oriental Magpie as a common species in Nanjing, and its widespread distribution reflects a strong capacity for environmental adaptation. Nevertheless, significant differences in its FID were observed among different park types ($P < 0.001$). Moreover, in the most developed sites, the goodness-of-fit of the XGBoost model for this species was lower than in other sites, suggesting that birds adjust their escape strategies flexibly in response to habitat modifications, highlighting their behavioral plasticity (Bar-Ziv et al., 2023; Vincze et al., 2019).

As a widely used behavioral metric, FID provides a measure of animals' responses to potential threats (Kalb et al., 2019; Ortiz-Jimenez et al., 2022). Traditionally, it has been applied in low-disturbance environments such as nature reserves (Braumohr et al., 2018) to assess the innate defensive thresholds of birds. By contrast, our study focuses on human-dominated urban parks, thereby providing empirical evidence of birds' behavioral adaptability in urban contexts and underscoring the unique value of FID in urban ecology. Because urban parks primarily serve local residents, balancing human recreational needs with the protection of bird habitats is a critical management challenge (Prestes et al., 2018). Our XGBoost results indicated that environmental variables differed in their influence on FID, suggesting that targeted interventions could help modulate birds' behavioral responses to human disturbance. The variation in escape distances across species not only reflects differences in risk perception but also reveals their contrasting capacities for urban adaptation. Thus, park planning and management should emphasize more on the preservation and optimization of refuge spaces, particularly ecological buffer zones for species with longer FID and higher sensitivity to disturbance (Hagy et al., 2017). In other words, instead of applying uniform management strategies, decisions should be guided by interspecific differences.

4.4. Research limitations and future directions

Our study primarily reflects the overall status of birds in urban parks, though unobserved deviations may exist in the behavioral patterns of different functional groups. In particular, migratory bird records accounted for only about 20 % of the total observations, which was insufficient to support an in-depth comparison between migratory and resident species. Migratory birds typically face more complex and diverse risks across habitats and exhibit higher vigilance (Morelli et al., 2022a). This partly explains why it is difficult to collect sufficient migratory bird data in environments with frequent human activity, such as urban parks. Moreover, the influence of group dynamics on individual FID varies significantly across species, and is better suited for detailed studies at the biological level (Linley et al., 2019; Zhao et al., 2024b). Future research could address these gaps by expanding the sample scope and extending observation periods within single seasons, thereby clarifying the behavioral differences of functional groups under urbanization.

Drawing on existing literature (Ardila-Villamizar et al., 2022; Lethlean et al., 2017) and considering field conditions, we ultimately selected two variables, "sound level" and "distance from paths and roads", to represent human disturbance at the sample sites, reflecting

short-term and long-term effects, respectively. Other indicators, such as visitor numbers, activity types, and paved area, were considered in the design phase (Bötsch et al., 2017; Mayo et al., 2015) but not adopted, as limited human presence in some sites made reliable data collection impractical. Furthermore, bird activity was relatively scarce in highly disturbed areas, which constrained the availability of FID data and reduced the representativeness of these variables. We also observed frequent bird feeding by humans in several urban parks, particularly around plazas. Previous studies have shown that long-term feeding can lead to dependence on humans, even altering birds' escape thresholds (Fleming and Bateman, 2017).

Overall, our study prioritized environmental factors with broader applicability and greater feasibility to uncover the general ecological responses of birds to urban disturbance. While this approach strengthened the robustness of the analysis, it also carried limitations. The multidimensional nature of human disturbance was not fully captured, potentially underestimating the impact of persistent local disturbances. In fact, practices such as human feeding and inter-individual variation among birds may indirectly shape behavioral trajectories through resource inputs, offering important avenues for future research. Building on our findings, subsequent studies should integrate more dynamic and spatially explicit disturbance variables to develop a more comprehensive indicator system and enhance mechanistic insights into human impacts. We further recommend targeted investigations at finer scales, particularly in highly developed urban parks, with expanded sampling and functional zoning approaches to generate more effective strategies for optimizing urban green space.

5. Conclusion

This study underscores FID as a crucial behavioral indicator of birds' stress responses and survival pressures. Variations in FID reflect individual risk perception as well as adaptive differences at the community level. Incorporating FID into urban ecological management provides valuable insights into how birds perceive and respond to human presence, while also offering a simple and sensitive tool to assess whether green spaces provide sufficient ecological security. At present, urban green spaces design often prioritizes landscape aesthetics and recreational use while overlooking the behavioral needs of wildlife. Future planning of green spaces should place greater emphasis on the behavioral heterogeneity of birds under varying conditions. This calls for an integrated approach that considers vegetation configuration, spatial layout, and patterns of human activity to develop more targeted conservation strategies. By fostering heterogeneous habitats with controllable levels of disturbance, urban green spaces can become more ecologically inclusive, promoting dynamic coexistence and reciprocal adaptation between human activities and urban bird communities.

CRedit authorship contribution statement

Zhewen Ye: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Qian Xiong:** Writing – review & editing, Visualization, Investigation, Data curation. **Yao Xiong:** Writing – review & editing, Supervision, Funding acquisition. **Yunfeng Yang:** Writing – review & editing, Supervision, Resources, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2025.114455>.

Data availability

Data will be made available on request.

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